

Proof. In the first case, we set $v_p = (\nabla X|_p)^{-1}w_p$ and apply the Cauchy-Schwarz inequality to obtain

$$\langle \nabla_{v_p} X, v_p \rangle = \langle w_p, (\nabla X|_p)^{-1}w_p \rangle \leq \|(\nabla X|_p)^{-1}\| \|w_p\|^2 = \|(\nabla X|_p)^{-1}\| \|\nabla_{v_p} X\|^2$$

so $-\sup_{p \in \mathcal{U}} \|(\nabla X|_p)^{-1}\|$ can be taken as an upper bound for α_p .

In the second case, we choose $v_p = r_p + \beta n_p$ where $r_p \in \text{Range}(\nabla X|_p^*)$, $\beta \in \mathbb{R}$ and $n_p \in \ker(\nabla X|_p) \neq 0$. Then, the condition (25) reads

$$\langle \nabla_{r_p} X, r_p \rangle + \beta \langle \nabla_{r_p} X, n_p \rangle \leq -\alpha \|\nabla_{r_p} X\|^2$$

where β is arbitrary. Therefore, it is necessary that $\langle \nabla_{r_p} X, n_p \rangle = 0$ which is true for all choices of r_p and n_p if and only if $\nabla_{r_p} X \in \text{Range}(\nabla X|_p^*)$. By definition, the set of all $\nabla_{r_p} X|_p$ is precisely $\text{Range}(\nabla X|_p)$, so we conclude that one must have $\text{Range}(\nabla X|_p) = \text{Range}(\nabla X|_p^*)$. The proof is completed in the same way as in the non-singular case, by restricting $\nabla X|_p$ to $\text{Range}(\nabla X|_p^*)$. $\square$

Note that this proposition does not ensure that the vector field X is cocoercive since it might happen (as in the bound of the proof) that $\alpha < 0$.

A coordinate formula for the cocoercivity constant. We give a concrete formula for the cocoercivity constant when it exists. Suppose we have introduced local coordinates $x_1, \dots, x_d$. Now g is to be interpreted as a $d \times d$ matrix with elements $g_{ij} = \langle \partial x_i, \partial x_j \rangle$. The matrix of the operator $\nabla X|_p$ is denoted $\mathcal{A}_{X,p}$. We assume that we can compute its reduced singular value decomposition $\mathcal{A}_{X,p} = U \Sigma V^T$ where $U, V \in \mathbb{R}^{d \times r}$ have orthonormal columns and $\Sigma \in \mathbb{R}^{r \times r}$ is diagonal and positive definite and where r is the rank of $\mathcal{A}_{X,p}$. Thanks to Proposition 12 we can write $U = VQ$ where Q is orthogonal and $r \times r$, i.e. $Q = V^T U$. We are interested in finding the largest possible α_p such that

$$v_p^T g \mathcal{A}_{X,p} v_p \leq -\alpha_p v_p^T \mathcal{A}_{X,p}^T g \mathcal{A}_{X,p} v_p, \quad \forall v_p \in \mathbb{R}^d$$

It is easy to show that $-\alpha_p$ can be chosen as the largest eigenvalue

$$-\alpha_p = \max_{1 \leq i \leq d} \lambda_i \left(\frac{M + M^T}{2} \right)$$

with

$$M = \tilde{g}^{-\frac{1}{2}} \Sigma^{-1} V^T U \tilde{g}^{\frac{1}{2}}, \quad \tilde{g} = U^T g U.$$

To find α it suffices to maximize α_p over the domain $\mathcal{U} \subset M$.

Formulas for μ_+ and μ_- in Theorems 6 and 9 can be derived similarly. In the case of ∇X non-singular, we obtain

$$\mu_+ = \lambda_{\max} \left(\frac{M_+ + M_+^T}{2} \right), \quad \mu_- = \lambda_{\max} \left(\frac{M_- + M_-^T}{2} \right),$$

where $\lambda_{\max}(A)$ denotes the maximum eigenvalue A and with

$$M_+ = -G^{1/2}(I - P_X)\nabla X^{-1}G^{-1/2}, \quad M_- = -G^{1/2}P_X\nabla X^{-1}G^{-1/2}.$$